Group 013: Perception Driven Robot Grasping

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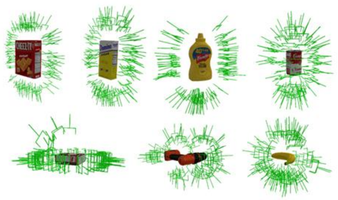
Motivation

- The need for robots to operate in unstructured environments is increasing across industries.
- Achieving reliable grasping of diverse objects is a key challenge for real-world deployment.
- Integrating object recognition and motion planning to enable more intelligent and adaptable robotic grasping.

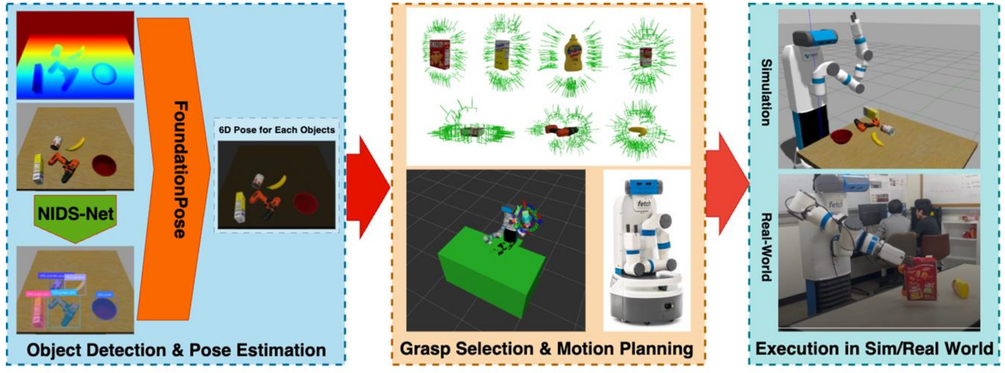
Overview

We developed a multimodal perception-driven pipeline for reliable grasp planning and execution in simulation and real-world scenarios, involving object detection, segmentation, and 6D pose estimation.

- **Object Detection & Segmentation:** NIDS-Net, a unified framework for Instance Detection and Segmentation.
- **Object Pose Estimation:** FoundationPose, a unified framework for 6D pose estimation.
- **Table-top Objects:** a subset of YCB-objects are selected to ensure the pipeline robustness across varied shapes and textures.
- **Motion Planning:** we use MoveIt to select the best grasp from the grasp dataset.
- **Environment setup:** we created the scene by placing objects randomly on table in Gazebo simulator and real world with Fetch robot.

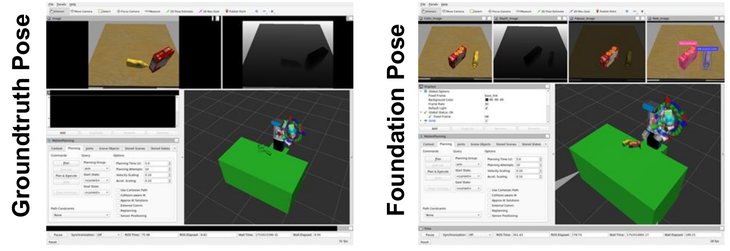


Technical Approach



Implementation on a Real Fetch Robot

Verification in the Simulation



Conclusions and Limitations

- Our proposed pipeline effectively integrates **NIDS-Net**, **FoundationPose**, and **MoveIt** to automatically detect objects, estimate their poses, and plan grasp motions.
- Verification results demonstrate that the **Fetch robot** can reliably perform pick-and-drop tasks in both simulated and real-world environments using this pipeline.
- **Limitations**: The pipeline depends on unified perception methods. However, in certain environmental conditions, objects may not be accurately recognized, leading to motion planning failures. Additionally, the object's physical dynamics can result in it being dropped during the grasping process.

Acknowledgments

We thank CAST-STEM, Dr. Yu Xiang, our TAs Jikai Wang and Qifan Zhang, and our CS Lab colleagues for their insightful guidance and valuable technical assistance during this research process.

Citations:

[1] Bowen Wen, Wei Yang, Jan Kautz, Stan Birchfield et al., "Foundationpose: Unified 6d pose estimation and tracking of novel objects," in CVPR, 2024.
[2] Yangxiao Lu, Yunhui Guo, Nicholas Ruozzi, Yu Xiang, et al. "Adapting pre-trained vision models for novel instance detection and segmentation," arXiv preprint arXiv:2405.17859, 2024.
[3] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo et al., "Segment anything," in ICCV, 2023.